

AskIt Research Lab: Home Assignment

Role: Analyst / Researcher, AskIt Research Lab

Time budget: 5 hours

Tooling: Claude Code, required. Github.

1. What this is

You will not finish this assignment. It is not scoped to be finished in five hours, and we know that. What you choose to do with the five hours, and what you consciously decide to drop, is a large part of what we are looking at.

We are not grading a result. We are trying to see how you think, see your creativity and problem solving. We also want to see how you decide what matters when you cannot do everything, and whether you can actually drive Claude Code rather than be driven by it.

2. How the lab works

The AskIt Research Lab has three functions. A researcher here touches all three most weeks, and this assignment puts you through all three so both sides can feel whether the fit is real.

Degrees of freedom and data. Deciding which dimensions of human variation we need to model at all, what data exists to measure ourselves against, and what the literature already knows. A question from this function: how does the way people *talk* about a decision relate to how they *behave* in an economic experiment about that decision, and where does the gap show up as a measurable bias?

Fast research. Roughly two week cycles, tied to product need. Something underperforms and the job is to close the gap quickly with whatever is defensible.

Slow research. Core capability. Persona architecture, calibration, steering vectors. Months, not weeks. Judged on whether it moves a ceiling, not on whether it ships this sprint.

3. The problem

An audience does not only believe different things. It sounds different.

Build a function:

$f(\text{audience description, persona description}) \rightarrow \text{text written by that person}$

The deliverable is the function, not its output. We're not looking for situational success that only works on certain types of audiences, but not the others. What we care about is how the output behaves as the inputs vary.

Please submit both the personas descriptions, and the audience description(see table for example of what those are). How you do that is part of the work and worth thinking about, because it is where variation between individuals inside an audience comes from.

Audience description versus persona description

An audience description defines a population. A persona description is one individual drawn from it. The audience carries ranges and distributions, the persona carries specific values.

Audience description	Persona description
US adults aged 18 to 24 who surf recreationally at least twice a month. Mostly coastal California. Mixed income, mixed education, mixed years of experience.	22, lives in Santa Cruz, works part time at a coffee shop while finishing community college, surfs three mornings a week before shifts, started at 12.

This is a format illustration only. It is not a claim about what a persona description should contain, how long it should be, or whether it should be free text or structured. Those are your decisions and we are interested in how you make them.

Requirements on the input space

- Work with at least four audiences.
- At least two must be a **test set**. You do not look at their output and do not tune against them until you have finished developing. They are there to tell you whether you built a function or a lookup table.
- At least one audience must be an intersection of two attributes rather than a single attribute.
- Choosing audiences - we prefer audiences you can actually get ground truth on is part of Part 1. If not, choose ones that you can hypothetically could get ground truth on.

Elicitation contexts

An elicitation context is the situation you put a persona in to produce text. For example: "tell me about your weekend."

Use the same contexts for every audience. If you ask one audience about their weekend and another about a cancelled subscription, the outputs will differ, but you will not know whether that is because the audiences differ or because the questions differ. Holding the context fixed means the only thing varying is the input you are studying.

Use at least two contexts, and choose them deliberately. Different contexts pull on the function in different ways, and testing only one kind will hide a real failure mode. In your writeup, say why you chose the contexts you chose and what each one was meant to reveal.

Data

Public data or research datasets only. If you can't find one, create a well thought synthetic dataset to use later as a stand in replacement. No login walls, no personal data, respect platform terms. You do not need much. A few hundred real texts per audience is enough for anything you will do in five hours. One clean number beats ten dirty ones. If your ground truth has a contamination problem, name it rather than working around it quietly.

4. The three parts

Part 1: Degrees of freedom and measurement

Before generating anything, decide what you are trying to move and how you would know if you moved it.

- Along which axes does the language of two audiences actually differ? Which are surface and which are structural?
- Which axes can you measure cheaply with data you can get today?
- What is your ground truth or improvement reference?
- You will need to report both how much output varies **within** an audience across personas, and how well audiences stay **separated** from each other. Decide now how you will measure each.
- Commit your metric before you run anything. We will compare it against what you report.
- What would a system that scores perfectly on your metric still get wrong?

Part 2: Fast research

Improve the function using the fastest defensible method. Prompt engineering is allowed.

- Establish a naive baseline first.
- Improve on it. At least one comparison must be quantitative.
- Report the within-audience and between-audience numbers separately. A system can look strong on one while failing the other.
- Run the held out audiences at the end, once. Report what happened, including if it was worse.
- Report at least one thing that did not work.
- Your samples will be tiny. State the uncertainty on your numbers, or state why you cannot.

Part 3: Slow research

Slow research is about finding a deeper solution to the problem that is higher risk but higher reward

Hard constraint: it cannot be a prompt engineering solution. If it can be implemented by changing the text sent to a model, it belongs in Part 2. Include the mechanism, why it should work, what it costs in data and compute and man hours, how you would falsify it, and the point at which you would kill it.

If you have time, run the smallest possible probe. A negative probe with a design that could have gone either way is worth more than no probe, and much more than a positive probe with a design that could not have failed. If needed, you are welcome to consider for this task a local LLM instead of LLM's like Claude or GPT.

5. Deliverables

1. **Repository.** Code plus data pointers. No large data files. Keep your real commit history, do not squash it at the end.
2. **Writeup(REPORT.md).** 1,500 words maximum. Your metric, what you did, your numbers,

what failed, the slow research proposal(and test), and what you would do with another week.

3. **One visual.** Baseline against best attempt, within-audience and between-audience, on developed and on held out audiences, with whatever uncertainty you can attach.
4. **Claude Code session artifacts.** Transcripts or session logs, dead ends included.
5. **Time log.** Rough is fine.
6. **One paragraph:** a specific place where Claude Code gave you something wrong, or plausible but wrong, how you caught it, and what you did about it. If it genuinely never happened, say that instead and we will take you at your word.

6. What we are looking at

No weights; we are not asking you to optimize an allocation. Roughly in order of how much each tells us:

- **Whether your measurement design captures the shape of the problem, or only its easiest part.**
- What you dropped, and whether you can defend the triage.
- Whether your numbers are honest, and whether the comparisons could have gone against you.
- **Whether the slow research proposal is a real mechanism with a stopping rule, or a slogan.**
- Whether you separate what you know from what you inferred from what you are guessing, and say which is which.
- **Whether the session log shows you steering the work, or accepting output.**

Not looked at: visual polish, slide design, volume, or whether the final result is good.

7. Practical

- Five hours. Stop and tell us where you stopped.
- Claude Code required.
- Any model API(or Claude Code agent spawn instead).
- Use any published research. Cite it.
- Questions are welcome and are not a penalty. The question you ask is information too.
- **Please make sure to commit changes as you work to github.**

8. After

A 30 minute conversation about your submission. We will push on your choices and disagree with some of them. Changing your mind under a good argument is a positive signal here. So is holding your ground under a bad one.

This assignment sits close to problems we actually work on.